



Bacteriocin producing microbes with bactericidal activity against multidrug resistant pathogens

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ARTICLE INFO

Article history:

Received 24 July 2021

Received in revised form 8 September 2021

Accepted 27 September 2021

Keywords:

Bacteriocin

Lactobacillus plantarum

Antibacterial resistance

Foodborne pathogens

ABSTRACT

Background: Bacteriocins are proteins or peptides synthesized by bacteria that show inhibitory or killing activities against various bacteria. Bacteriocins are mainly considered for effective alternatives to different commercial antibiotics, preservatives in the food and pharmaceutical industries.

Objectives: To screen and analyze novel bacteriocin-producing bacteria from the fermented food shidal for antibacterial activity against food pathogens and their molecular interactions studied through computationally.

Methods: In this study, a strain *Lactobacillus plantarum* LA21 was isolated from the fermented food shidal identified based on morphological, biochemical, and 16S rDNA gene sequencing. The potent bacterium was subjected to improve bacteriocins production and characterized. Antimicrobial activity against drug-resistant bacteria and minimum inhibitory concentration (MIC) were determined. The bacteriocin was treated with proteolytic enzymes, and the mechanism of action on food pathogens was analyzed. Molecular docking studies were carried out as GLIDE module in the maestro tool of Schrodinger Software.

Results: Bacteriocin was effective against pathogens such as *Bacillus pumilus*, *Bacillus amyloliquefaciens*, *Staphylococcus aureus*, and *Listeria monocytogenes*, with the most negligible MIC value was detected in *L. monocytogenes*. Furthermore, the depleted viability of bacterial cells indicated bacteriocin-induced cell lysis in *L. monocytogenes* via bactericidal activity. In addition, proteolytic enzyme digested bacteriocins revealed bacteriocin-like substances. Finally, molecular docking was performed to study the interactions between the targets and bacteriocins, results in relative intense contact with minimally 3 Å distance.

Conclusions: The characteristic features of these bacteriocin-like molecules revealed that *L. plantarum* LA21 is a novel bacteriocin-producing bacterial strain to prepare novel antimicrobial drugs, feed additives or preservatives for future use in livestock and food industries.

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Introduction

Lactic acid bacteria (LAB) are widely used in food preservation and enhance the nutrient level of various fermented foods and products [1]. The LAB produces lactic acid, which changed the environment into more acidic, reduces the pathogenic microbial population. Also, LAB has various low molecular weight substances,

including bacteriocins, diacetyl (2,3-butanedione), carbon dioxide, hydrogen peroxide [2], and these compounds as capable of inhibiting growth of various pathogenic bacteria [3]. In recent years, people show much attention to applying LAB as a good additive in the fish food industry due to its generally recognized as safe (GRAS) status and antimicrobial properties [4–6]. Food preservation techniques usually can prevent food contamination; moreover, various pathogenic strains have been reported from the bio-preserved food [7]. Furthermore, the foodborne pathogens show increased resistance against multiple antibiotics available commercially [8]. These antibiotic resistance patterns among bacteria have generated more interest in the analysis of any novel class

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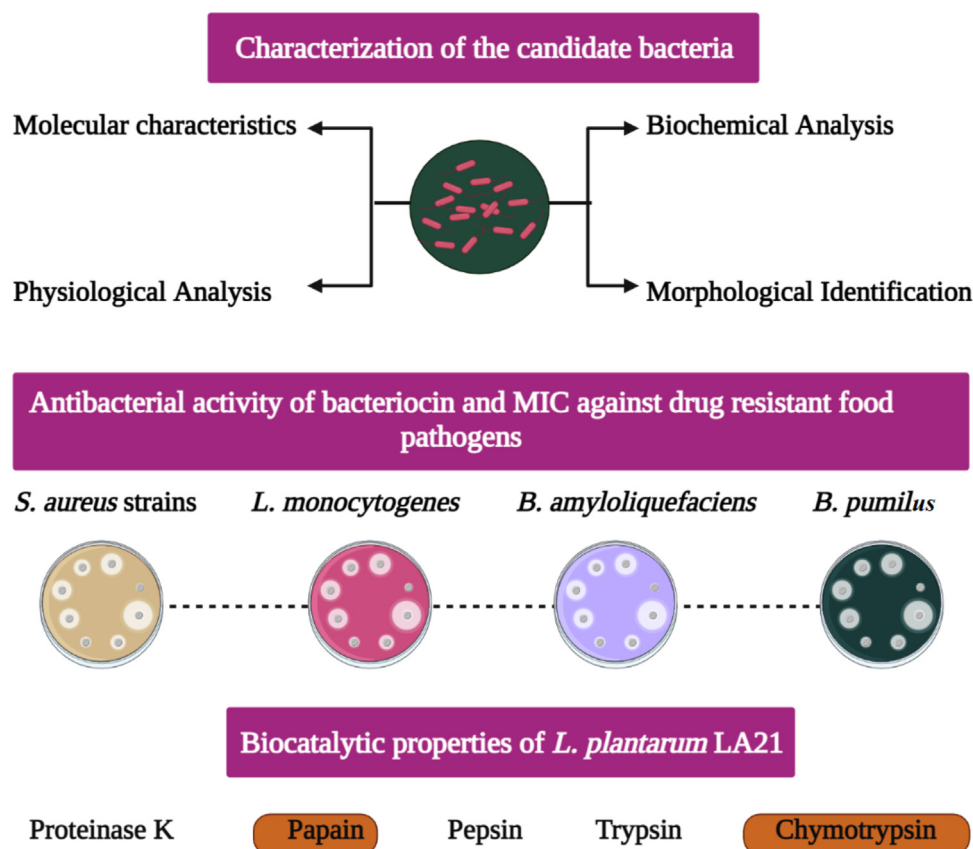


Fig. 1. Schematic representation of present study.

of antibiotics/secondary metabolites from natural sources capable of treating various pathogenic bacterial strains [9–11].

In recent years, consumers prefer various natural food preservatives instead of synthetic additives. These natural food preservatives have been widely used to reduce the usage of commercially available artificial additives. Administration of LAB in food products reduces the pathogenic microorganism in the food products [12]. In recent years, various studies showed that the efficacy of LAB to affect the growth of food pathogens [8]. Bacteriocins produced by LAB are beneficial to the human body and inhibit the growth of various foodborne pathogenic microorganisms such as *Staphylococcus aureus*, *Escherichia coli*, and *Listeria monocytogenes* [13,14]. Therefore, the bacteriocins have been used widely as natural food preservative agents. Bacteriocin-producing microorganisms have been screened from various natural sources, including fermented food to produce bacteriocins [15]. The organisms such as *Lactobacillus plantarum* PMU 33, *Enterococcus faecium* NKR-5-3, *Pediococcus pentosaceus* CFF4, and *E. faecium* CN-25 [16–19] have been isolated from the fermented food products for the production of bacteriocins.

The traditional drug discovery approach is complicated, difficult, expensive, time-consuming, and laborious [20]. Computational methods such as molecular docking have played a critical role in simplifying the road to drug discovery in order to overcome these obstacles. A molecular docking technology is currently a helpful drug development tool for screening candidates from the drug libraries quickly. In this study, a novel bacteriocin-producing *L. plantarum* SAL-03 isolated from fermented fish was characterized. In addition, the mode of action of bacteriocin, physicochemical characteristics were also analyzed for food preservative applications. Furthermore, the novel bacteriocin was evaluated *insilico* docking by using GLIDE docking in Maestro, Schrodinger software.

Materials and methods

Bacteria and culture condition

The fermented food product shidal was collected from the North-Eastern part of India. The LAB from the shidal were cultured using a MRS medium (HiMedia, India). The sample was serially diluted and cultured in an MRS broth medium for 24 h. Tryptone soy yeast extract broth was used for the culture of *L. monocytogenes* 54002. It was stored at -80°C with 20% (v/v) glycerol and used as an indicator organism.

Screening of LAB for the production of bacteriocin

The LAB enriched from the samples were cultured on MRS agar plates. Samples were serially diluted and grown for 24–48 h at 37°C . A total of 49 isolated bacterial strains were incubated into individual vials containing 10 mL MRS broth medium for 24 h. The culture was centrifuged at 10,000 rpm for 10 min, and the pH of the supernatant was adjusted to 6.8 ± 0.2 using NaOH (1 M). The culture supernatant was filtered using a $0.22\ \mu\text{m}$ filter, and antimicrobial properties were detected using the agar well diffusion analysis method. The antibacterial activity of the cell-free extract was tested against indicator bacterial strain, and the zone of inhibition (mm) was analyzed. The tested bacterial isolates (49 isolates) with maximum activity were tested against food pathogens such as *Bacillus pumilus*, *Bacillus amyloliquefaciens*, *S. aureus*, and *L. monocytogenes*. The culture plates were incubated at 37°C for 24 h, and the zone of inhibition around the well was tested. The bacterial isolate, which showed a maximum inhibition zone, was used for subsequent analysis.

Table 1
Growth and antibacterial activity of cell-free extract from *L. plantarum* LA21.

Incubation time (h)	Biomass Absorbance at 600 nm	Zone of inhibition (mm)	pH of the medium
6	0.32 ± 0.02	0	6.8 ± 0.2
12	0.48 ± 0.07	2 ± 1	6.7 ± 0.01
18	0.95 ± 0.21	14 ± 2	6.3 ± 0.01
24	1.13 ± 0.28	16 ± 1	6.2 ± 0.04
30	1.97 ± 0.13	21 ± 2	5.3 ± 0.16
36	2.18 ± 0.42	19 ± 2	4.8 ± 0.22
42	2.26 ± 0.3	16 ± 1	4.6 ± 0.15
48	2.27 ± 0.29	15 ± 3	4.53 ± 0.08

Table 2
Antibacterial activity of bacteriocin and minimum inhibitory concentration against drug resistant food pathogens.

Antimicrobial activity	Zone of inhibition (mm)	MIC (μg/mL)
<i>S. aureus</i> strain K2	13 ± 1	30
<i>S. aureus</i> strain L4	15 ± 1	15
<i>S. aureus</i> strain M2	9 ± 2	30
<i>L. monocytogenes</i>	24 ± 2	15
<i>B. amyloliquefaciens</i> CD2	19 ± 2	15
<i>B. amyloliquefaciens</i> CD7	17 ± 1	30
<i>B. amyloliquefaciens</i> CX3	20 ± 3	15
<i>B. pumilus</i> X1	19 ± 1	30
<i>B. pumilus</i> - X2	10 ± 2	30
<i>B. pumilus</i> LD7	8 ± 1	30

Characterization of bacteriocin-producing strain

The selected bacterial strain LA21 was characterized based on carbohydrate-fermentation, catalase-reaction, morphological, biochemical properties and 16S rDNA gene sequencing. The genomic DNA of the bacterial strain was purified using a DNA extraction kit based on manufactures instructions. The 16S rDNA of the organism was amplified using universal forward (5'GAGTTTGATCMTGGCTAG3') and reverse (5'ACGGGCGG TGTGTRC3') primer, and GenBank accession number was assigned. The schematic presentation of work plan was depicted in Fig. 1.

Growth kinetics and antibacterial property of bacteriocin from *L. plantarum* LA21

The strain *L. plantarum* LA21 was cultured in a MRS broth medium (50 mL) in a 250 mL Erlenmeyer flask until the absorbance of the bacterial cells reaches 0.5 OD (at 600 nm). Then the culture was inoculated in a 1.0 L capacity Erlenmeyer flask containing an 800 mL culture medium. Then it was incubated for 72 h, and the growth kinetics was determined. The bacterial culture was withdrawn every 6 h, and the results were described. The antimicrobial activity of bacteriocin was tested by the agar-well diffusion method and expressed as zone of inhibition (mm).

The action of bacteriocin on foodborne pathogen

To analyze the mode of action of bacteriocin extracted from *L. plantarum* LA21, the indicator strain was used to test the mode of action. The indicator strain, *L. monocytogenes* was cultured in an Erlenmeyer flask containing 100 mL culture medium. It was incubated at 37 °C for 24 h. The growth was monitored at 600 nm using a UV-vis spectrophotometer, and the cells were harvested by centrifugation (10,000 rpm, 10 min). Then it was resuspended in phosphate buffer saline (pH 7.2, 0.1 M), and the cell concentration was approximately 1×10^8 CFU/mL. The crude bacteriocin was added to the culture at 300 mg/L concentration in the experimental

Table 3a
Effect of enzymes on cell-free extract of *L. plantarum* LA21.

Enzyme	Zone of inhibition (mm)	pH
Proteinase K	0	7.4
Papain	3 ± 1	6.6
Pepsin	0	3.2
Trypsin	0	7.5
Chymotrypsin	2 ± 1	7.9
Control	22.4 ± 0.3	7.2

Table 3b
Effect of pH on antibacterial activity of the extract from *L. plantarum* LA21.

pH	Zone of inhibition (mm)
3.5	10 ± 2
4	14 ± 1
4.5	14 ± 3
5	15 ± 1
5.5	15 ± 2
6	18 ± 1
6.5	19 ± 2
7	20 ± 1
7.5	20 ± 2
8	15 ± 1
8.5	10 ± 2
9	10 ± 3
9.5	8 ± 1

trial. The culture was incubated for 3 h at 37 °C, and the cell viability was observed every 30 min. A nutrient agar plate was used for the determination of viable bacterial cells. The well-developed bacterial colonies were counted using an automated colony counter. A sample was not added to the control, and an experiment was performed in the same assay condition.

Effect of culture supernatant on potassium ion efflux

The influence of cell-free extract of *L. plantarum* LA21 on the efflux of potassium ion from *L. monocytogenes* was evaluated. The amount of free potassium ion in *L. monocytogenes* was considered after exposing the culture to the cell-free extract of *L. plantarum* LA21 at 1X MICs for 0, 20, 40, 60, 80, 100, 120, 140, 160, and 180 min. Potassium concentration was determined at these time intervals, and the results were expressed as mmol/L free potassium in the culture media.

Effect of enzymes on bacteriocin extracted from *L. plantarum* LA21

The effect of partially purified bacteriocin on various enzymes was tested by incubating 200 mg bacteriocin mixed with different proteolytic enzymes at optimal pH value. After 3 h incubation with proteases, the remaining bacteriocin activity was analyzed. To the control, enzymatic treatment was not subjected. Then the sample was subjected to test antagonistic potential against indicator strain.

Effect of temperature and pH on bacteriocin activity

The pH stability of bacteriocin from the strain *L. plantarum* LA21 was evaluated by adjusting the pH of the sample using 1 N NaOH or 1 N HCl. After 3 h incubation, the remaining activity was analyzed. To analyze the thermal stability of bacteriocin, the crude sample was incubated at various temperatures for 1 h. The residual enzyme activity was assayed, and the sample maintained at 30 ± 2 °C was used as a control.

Table 4

Effect of cell free extract of *L. plantarum* LA21 on the viability of the indicator bacterium.

Min	Log Cell viability (CFU/mL)	
	Cell free extract	Control
0	8.32 ± 0.14	8.3 ± 0.41
30	7.9 ± 1.3	8.34 ± 0.38
60	5.8 ± 1.1	8.4 ± 0.42
90	5.7 ± 0.3	8.5 ± 0.34
120	2.4 ± 0.5	8.6 ± 0.41
150	1.5 ± 0.3	8.7 ± 0.32
180	0 ± 0	8.81 ± 0.24

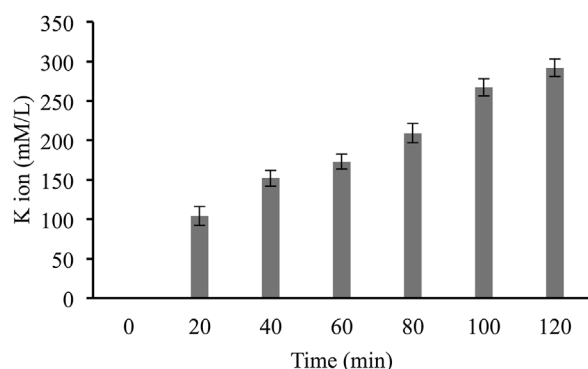


Fig. 2. Effect of cell-free extract of *L. plantarum* LA21 on potassium ions leakage from the indicator bacterium.

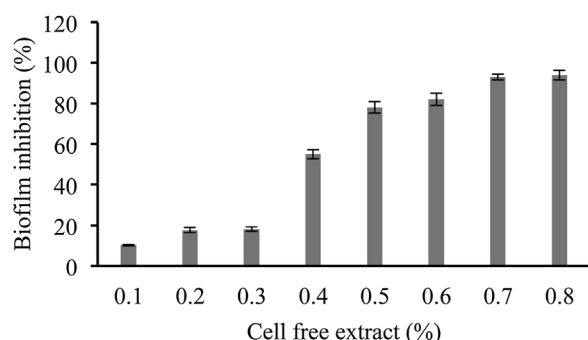


Fig. 3. Biofilm inhibitory activity of cell free extract isolated from *L. plantarum* LA21.

The role of the cell-free extract on control of biofilm development on food pathogen

The *L. monocytogenes* culture was inoculated in a nutrient broth medium, and biofilm inhibition ability was analyzed using the cell-free extract of *L. plantarum* LA21. About 3 mL of *L. monocytogenes* culture was grown in a nutrient broth medium, and cell-free extract (1X MIC) was mixed with pathogenic bacterial strain. The blank bacterial strain was not inoculated, and all experiments were performed for two days at 28 ± 2 °C. After 48 h, the tubes were washed two times with double distilled water, and the attached bacterial cells were fixed with methanol for 20 min and dried to obtain constant weight. In addition, to the developed biofilm, crystal violet dye (2%) was flooded and incubated for 10 min. The optical density of the attached bacterial cells released from the vials was measured at 595 nm against blank.

Protein preparation

The X-ray crystallography structure of surface proteins (PDB ID: 4JDZ) and transcriptional regulator proteins (PDB ID: 3MLF)

Table 5

Amino acid interaction with the bond length profile of two main targets.

PDB	Amino acid interaction (3-Letter code)	Bond length Å
3MLF	ASN7, ILE58, PHE59, LEU60,	1.86, 2.31, 2.57, (2.19, 1.92)
4JDZ	ASN257, ASN 381, LYS321, HIS322, ALA320, THR379,	2.39, 2.66, 2.10, 2.37, (2.22, 1.98), 2.25

were retrieved from the database of Protein Data Bank (Fig. 4). The preprocess of protein were carried in Protein preparation wizard of Maestro platform, protein structure following assigning bond orders, adding hydrogens, creating zero-order bonds to metals and disulfidebonds, converting selenomethionines to methionines and filling missing side chains using prime, inhibitor present in the crystallized protein was removed, and the refinement of H-bond optimized and finally applied energy minimization by an OPLS4 force field (Schrödinger Release 2021-3: Protein Preparation Wizard; Epik, Schrödinger, LLC, New York, NY, 2021; Impact, Schrödinger, LLC, New York, NY; Prime, Schrödinger, LLC, New York, NY, 2021).

Ligand preparation

The ligand was retrieved from the PubChem database (Compound CID: 17933559) in Structure Data File (SDF) (Fig. 5) and computationally generating 32 stereoisomers by applying OPLS4 force field energy minimization using the LigPrep module in the Maestro platform. Tautomers were developed, and conformations of the orientation of the ligand were carried out for further analysis (Schrödinger Release 2021-3: LigPrep, Schrödinger, LLC, New York, NY, 2021).

Molecular docking

The extra precision (XP) mode of the GLIDE program was used for all docking calculations. The docked ligand and protein interaction was analyzed by best pose, viewed in a pose viewer for best conformation. 2D interaction of protein and ligand were retrieved from ligand interaction module (Schrödinger Release 2021-3: Glide, Schrödinger, LLC, New York, NY, 2021).

Results

Screening and analysis of LAB for the production of bacteriocin

A total of 126 bacterial strains were isolated from shidal after enrichment culture in MRS broth medium. Among the bacterial strains, 49 bacteria showed antibacterial activity against indicator organisms. After primary and secondary screening, only one bacterial strain (LA21) was selected for further studies.

Characterization of bacterial strain

The selected strain was identified by biochemical and morphological methods and characterized as *L. plantarum* LA21. It was a rod-shaped bacterium, Gram-positive, facultative anaerobic strain, no spores found within the cell, and catalase-negative. It grew between 20 and 40 °C, and growth was high at 3.5% NaCl concentration. The selected bacterial strain metabolized lactose, glucose, galactose, fructose, xylose, mannose, sucrose, and maltose.

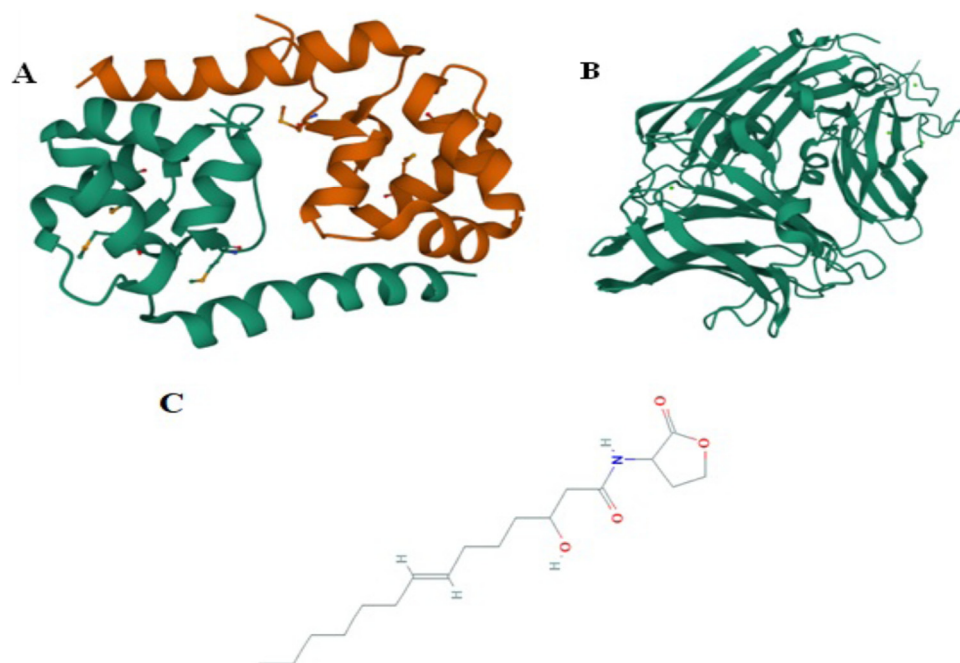


Fig. 4. Three-dimensional structure of transcriptional regulators proteins (A) and surface proteins (B), chemical structure of bacteriocin (C).

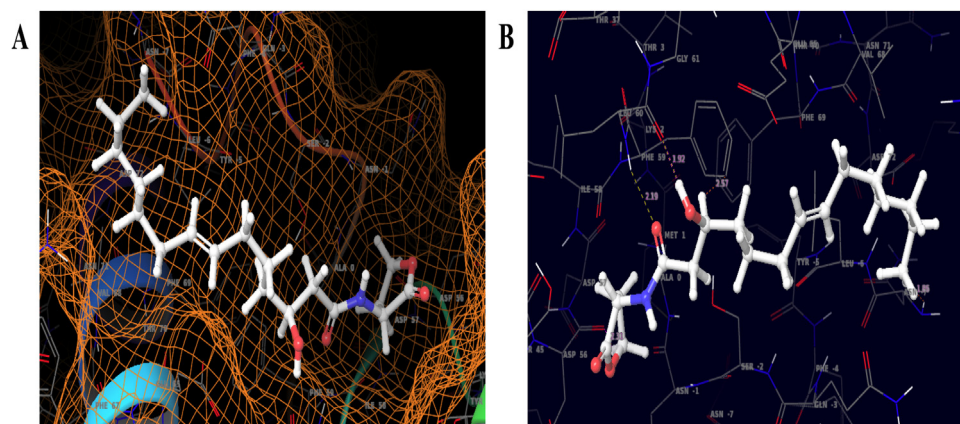


Fig. 5. The cavity site in mesh surface of bacteriocin with transcriptional regulator (A), the three-dimensional structure of the complex (B).

Growth kinetics of bacteriocin producing *L. plantarum* LA21 isolated from the fermented food

The growth profile of *L. plantarum* LA21, pH of the medium, and bacteriocin production were determined, and the results are described in Table 1. The growth was maximum after 24 h incubation, and bacteriocin production was maximum after 36 h and reached stationary phase after 42 h incubation. Bacteriocin production has been improved during the stationary growth phase of the organism. The pH of the culture medium declined gradually from 6.8 ± 0.2 to 4.53 ± 0.08 after two days of fermentation.

The absorbance of the culture was measured at 600 nm. The zone of inhibition was as measured in mm and the result was expressed as mean $\pm$ standard deviation.

Antimicrobial activity of bacteriocin from *L. plantarum* LA21

The antibacterial activity of cell-free extract against the indicator organisms, *B. pumilus*, *B. amyloliquefaciens*, *S. aureus*, and

L. monocytogenes were tested. *L. plantarum* LA21 exhibited more antibacterial activity against *L. monocytogenes*, than *S. aureus* and *B. amyloliquefaciens*. The MRS medium was used for the culture of the probiotic strain, and the MIC value was calculated. Further, the result was depicted in Table 2.

The zone of inhibition was expressed as mean $\pm$ standard deviation.

Biocatalytic properties of *L. plantarum* LA21

The bacteriocin treated with protease enzymes such as, trypsin, proteinase K, papain, and pepsin affected antimicrobial activity (Table 3a). pH is one of the essential factors determining bacteriocin activity. The bacteriocin was widely active between pH 3.5 and 7.5, and maximum activity was observed at neutral pH. Moreover, the inhibitory activity was only 8 ± 1 mm after being treated at pH 9.5 (Table 3b).

Bacteriocin was incubated with proteases at optimized pH value and the result was expressed as mean $\pm$ standard deviation.

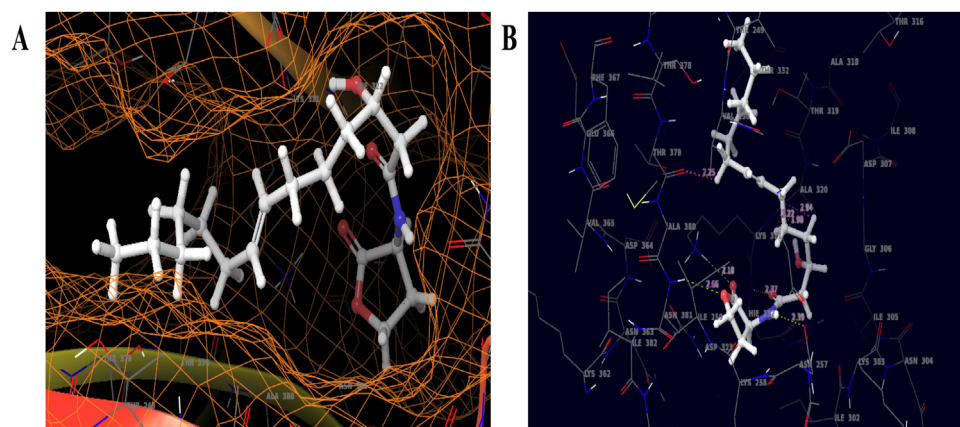


Fig. 6. The cavity site of in mesh surface of bacteriocin with surface proteins (A), the three-dimensional structure of the complex (B).

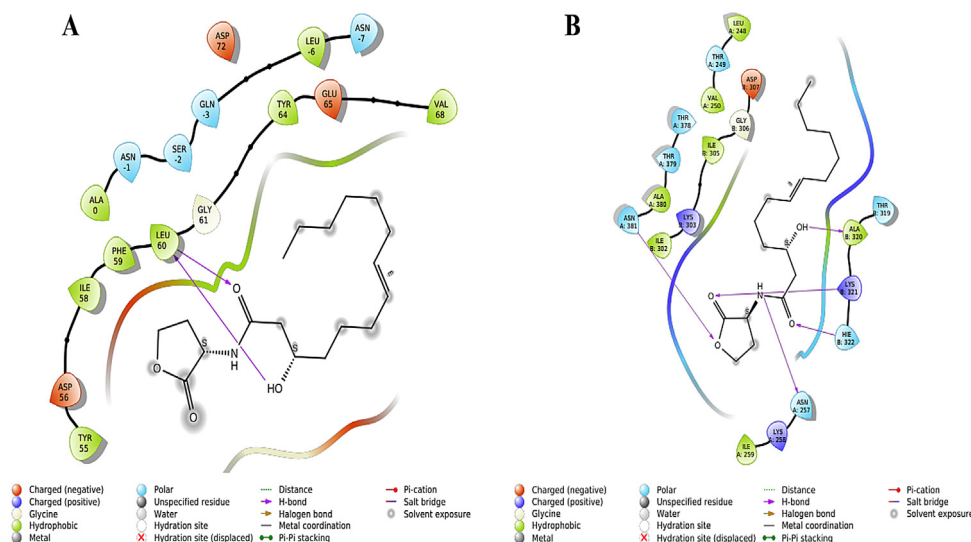


Fig. 7. Two dimensional structure of bacteriocin with transcriptional regulator (A) and surface proteins (B).

Bactericidal activity

The indicator strain was treated with bacteriocin at various incubation times. The addition of bacteriocin inhibited the growth of bacteria, and the bacterial population was depleted. The viable cells were not detected after 3 h incubation. The depleted viability of bacterial cells indicated bacteriocin induced cell lysis in *L. monocytogenes* via bactericidal activity. Initially, the cell-free extract did not affect the cell viability up to 30 min and later inhibited the growth and multiplication process. After 3 h incubation with cell-free extract, complete inhibition was observed (Table 4).

Influence of potassium ion leakage

The cell-free extract of *L. plantarum* LA21 induced the release of K^+ ions from the pathogenic bacteria (Fig. 2). The culture medium supplemented with cell-free extract at 1X MIC resulted in rapid release of K^+ ions from the bacterial cell. In the control experiment, no K^+ ions leakage was observed.

Biofilm inhibitory effect of cell-free extract

The impact of cell-free extract of *L. plantarum* LA21 on biofilm inhibition was tested using crystal violet staining, which revealed a significant inhibitory effect of the cell-free extract of LA21 on

L. monocytogenes. Furthermore, the cell-free extract induced inhibition of biofilm synthesis in bacteria, and the control (without cell-free extract) biofilm development was observed (Fig. 3).

Molecular docking

The 3D structure of transcriptional regulators proteins and surface proteins were depicted in Fig. 4A and B. The chemical structure of bacteriocin was retrieved from Chem database and illustrated in Fig. 4C.

The molecular docking was performed with two different targets of surface proteins and transcriptional regulation protein against bacteriocin using GLIDE, which was generally ranked accordingly by score calculation by the program in the form of G-Score. The lowest energy posture (binding confirmation) predicted by the object scoring function is used to determine a docking procedure's accuracy. The XP GLIDE docking approach was confirmed, and the G-score, GLIDE energy, and G-score of two different target proteins were evaluated. The amino acid interaction with their bond length was represented in Table 5. The docking complex results that, around 3 Å distance is the standard range of hydrogen bond. In our studies, two main complex results are below the range, making the interaction stronger among the complex. The ligand molecule was docked, and occupancy was highly interactive,

and their 3D (Figs. 5 and 6) and 2D interaction diagram was listed in Fig. 7.

Discussion

In the present study, the LAB were isolated from the fermented food products and the potent strain was characterized as *L. plantarum* LA21 based on their biochemical, morphological, physiological, and molecular characteristics [21]. The isolated LAB strains showed that the remarkable activities on various food-borne bacterial pathogens and confirmed by agar well-diffusion assay using indicator bacteria. The isolated LAB have been showed a significant antimicrobial activity against selected foodborne bacterial pathogens [22]. In addition, the LAB have been active against various foodborne pathogenic strains [23–25]. In subsequent experiments, *L. plantarum* LA21 was selected to analyze the antimicrobial property of cell-free extract. This study reveals that the exposure to cell-free extract of LA21 reduces the viability of bacterial cells. Recently, Bajpai et al. [26] reported that the inhibitory effect of cell-free extracts of LAB isolated from the Korean fermented food, Kimchi. Carina Audisio et al. [27] reported the antibacterial impact of LAB against various foodborne pathogens and inhibited cell viabilities.

The cell membrane of bacteria plays a critical role in various cellular functions, including the transport of electrolytes such as potassium ion, which participates in multiple pathways and is involved in enzyme functions. In bacteria, electrolytes regulation is critical because elevated leakage of ions can cause membrane damage and maintaining the balanced state of ions is essential to sustain survival and energy status [28]. Variations in the bacterial cell membrane structure may elevate the level of potassium ion release and affect cell metabolism, thus cause cell death [29]. In LAB, antimicrobial properties may be highly related to the involvement of various compounds, such as bacteriocin-like substances, H₂O₂, and organic acids [29]. Previous studies revealed that the antimicrobial potential of bacteriocin-like substances or bacteriocins against types of bacteria [30]. Enzymatic treatment of trypsin and other enzymes affected antimicrobial activity, showing the antibacterial activity of *L. plantarum* LA21 might be due to the production of bacteriocin or bacteriocin-like substances as described previously by Ghrairi et al. [31]. The present finding indicated that the inhibitory effect of cell-free extract of LA21 might be due to the production of organic acids. The cell-free extract was neutralized with acid and showed a very least activity indicated organic acid synthesis by the strain LA21. Pereira and Gomez [32] have been reported that the antibacterial potential of *L. acidophilus* against *S. aureus* and *E. coli*. The biosynthesis of organic acids effectively contributes to the control the growth of pathogenic microbes.

The derivatives of bacteriocin explored in antiviral activity against of Hepatitis E, targeting capsid protein, and furthermore their antimicrobial activity was studied from *in vivo* analysis in porcine and swine production [33–35]. In our studies, the potential antagonist activity of bacteriocins was seen strong interactions in surface proteins than a transcriptional regulator protein. Finally, the bond length of their interaction was notably under 3 Å distance which showed that the promising drug candidate against the disease targets.

Conclusions

In conclusion, bacteriocin-like molecules screened from *L. plantarum* LA21 and inhibited the growth of various food pathogens. Furthermore, the characterized bacteriocin was sensitive to various proteases and showed stability at different pH ranges. Therefore, it can be used as a feed additive, preservative in the livestock

agricultural and food industry as a novel antibacterial drug and food preservative. In addition, our potential drug candidate efficiency was also subjected to *in silico* molecular docking simulation, which reveals that the molecular interactions were seen promisingly, especially hydrogen bond contacts. The reported compounds need further validation of the *in vivo* model for potential candidates in clinical phase studies.

Funding

No funding sources.

Ethical approval

Not required.

Declaration of Competing Interest

The authors declare no potential conflict of interest.

Acknowledgments

The authors extend their appreciation to the Researchers supporting project number (RSP-2021/185), King Saud University, Riyadh, Saudi Arabia. The authors are grateful to their authorities for supporting this study and providing technical facilities for the work.

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